ModelIt!

Teacher's Guide

Trophic Cascades: When Predators Disappear

How Removing One Species Can Collapse an Entire Ecosystem

Quick Reference

Grade Level	Grade 08L2
NGSS Standard	MS-LS2-1, MS-LS2-4
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

© 2026 Discovery Collective. All rights reserved. For teacher use only.

Table of Contents

ModelIt!

1 NGSS Standards Unpacking & CAST Alignment 3

2 Scoring Rubric 4

3 Background Content Knowledge 5

4 The LEVER Framework 6

5 Slide-by-Slide Facilitation Guide 7

6 Answer Key & Expected Responses 9

7 Collaborative Learning Protocols 10

8 Differentiation & Extensions 11

9 Common Misconceptions 12

1 NGSS Standards & CAST Alignment

ModelIt!

Performance Expectation: MS-LS2-1, MS-LS2-4

Analyze and interpret data to provide evidence for the effects of resource availability on organisms and populations of organisms in an ecosystem. Construct an argument supported by empirical evidence that changes to physical or biological components of an ecosystem affect populations.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students build a computational model showing how changes to one trophic level cascade through the entire ecosystem.
Disciplinary Core Idea	**LS2.C Ecosystem Dynamics, Functioning, and Resilience** — Ecosystems are dynamic; changes to physical or biological components can cascade through the entire system.
Crosscutting Concept	**Cause and Effect** — Students trace the causal chain from predator removal through herbivore explosion, vegetation loss, erosion, and ecosystem instability.

CAST Testing Alignment

MULTIPLE CHOICE:

Wolves are removed from a forest ecosystem. Over the next 10 years, streambank erosion increases by 300%. Which chain of events best explains this?

CONSTRUCTED RESPONSE:

Using evidence from your model, construct an argument explaining why removing apex predators from an ecosystem can cause damage to components that seem completely unrelated to predators, such as soil and rivers.

2 Scoring Rubric

ModelIt!

Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

ModelIt!

For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Trophic cascades are among the most dramatic demonstrations of ecological interconnection. The removal or addition of a single species at the top of a food web can trigger a chain of effects that transforms entire landscapes — even reshaping rivers and coastlines.

What Is a Trophic Cascade?

A trophic cascade occurs when changes at one level of a food web ripple through other levels. Top-down cascades happen when predator changes affect everything below them. Bottom-up cascades happen when changes to producers (plants) affect everything above. The most dramatic examples involve top-down cascades triggered by predator removal or reintroduction.

The Yellowstone Wolf Story

Wolves were eliminated from Yellowstone National Park by 1926. Without predators, elk populations exploded and overgrazed willows, aspens, and cottonwoods — especially along riverbanks. Without root systems to stabilize banks, rivers widened and became shallower. When 41 wolves were reintroduced in 1995-96, elk behavior changed immediately. Elk avoided lingering near rivers (the 'landscape of fear'), vegetation regrew, banks stabilized, and rivers literally changed course. Beaver, songbirds, and fish populations recovered.

The Ecology of Fear

Predators affect prey not just by killing them but by changing their behavior. The 'landscape of fear' describes how prey avoid areas where predators hunt, allowing vegetation to recover in those zones. This behavioral effect can be as powerful as direct predation in shaping ecosystems.

Marine Trophic Cascades

Similar cascades occur in the ocean. When sea otters were hunted to near-extinction, sea urchin populations exploded and devoured kelp forests. Without kelp, entire coastal ecosystems collapsed. Restoring otter populations allowed kelp forests to recover, bringing back fish, invertebrates, and seabirds.

4 The LEVER Framework

ModelIt!

About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify apex predator population, herbivore population, vegetation health, erosion rate, and ecosystem stability as system components.

E **Establish Relationships**
Students determine that predators negatively affect herbivores, herbivores negatively affect vegetation, vegetation negatively affects erosion, and both vegetation and erosion affect ecosystem stability.

V **Visualize & Model**
Students build the trophic cascade model showing how predator removal triggers a chain reaction through the entire ecosystem.

E **Evaluate Outcomes**
Students compare balanced, predator-removed, and predator-reintroduced scenarios to measure the cascade effect and recovery time.

R **Revise & Extend**
Students add stream health or bird diversity components to extend the cascade model further.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

ModelIt!

Slide 1: Cover

2 min

Say: *"Same place, 20 years apart. What could possibly cause this much damage?"*

Watch For

Show Yellowstone before/after wolf removal photos. Let students react.

Slide 2: Learning Objectives

3 min

Say: *"Today we model how removing ONE species can collapse an entire ecosystem."*

Watch For

Pre-teach 'trophic cascade' with a domino metaphor.

Slide 3: Big Question

3 min

Say: *"You might think removing a dangerous predator makes things better. Think again."*

Watch For

Quick poll: Who thinks removing wolves would help or hurt Yellowstone?

Slide 4: LEVER Framework

2 min

Say: *"We'll build a model to trace the chain reaction from predators to rivers."*

Watch For

Preview the surprising connection between wolves and rivers.

Slide 5: Activity 1: Components

8 min

Say: *"What are the key parts of this ecosystem? Can you guess which ones wolves affect?"*

Watch For

Guide sorting. Build anticipation for indirect effects.

Slide 6: Activity 2: Connections

8 min

Say: *"Watch how each connection creates a domino effect through the whole system."*

Watch For

Map the chain step by step. Emphasize that wolves connect to rivers through FOUR links.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's remove the wolves and watch. Then let's bring them back."*

Watch For

Students predict each stage. Run removal, then reintroduction. Compare timelines.

Slide 8: Discoveries

5 min

Say: *"Wolves changed the course of rivers. Let that sink in."*

Watch For

Show the 'How Wolves Change Rivers' video summary. Discuss ecosystem interconnection.

Slide 9: STEM Challenge

12 min

Say: *"A park is in trouble. You're the ecologists hired to fix it. What does your model tell you?"*

Watch For

Teams design management plans using model evidence. Present and defend.

6 Answer Key & Expected Responses

ModelIt!

Component Sorting

- **EXTERNAL:** Apex Predator Population
- **INTERNAL:** Herbivore Population, Vegetation Health, Erosion Rate, Ecosystem Stability

Relationships

- **Apex Predator Population to Herbivore Population:** NEGATIVE (-) — More predators kill more herbivores, keeping their numbers in check. Fewer predators allow herbivore populations to explode.
- **Herbivore Population to Vegetation Health:** NEGATIVE (-) — More herbivores consume more plants, leading to overgrazing and vegetation decline.
- **Vegetation Health to Erosion Rate:** NEGATIVE (-) — Healthy vegetation holds soil in place with root systems. Less vegetation means more bare soil exposed to wind and water erosion.
- **Vegetation Health to Ecosystem Stability:** POSITIVE (+) — Healthy vegetation supports habitat, food, and shelter for diverse species, increasing overall ecosystem stability.
- **Erosion Rate to Ecosystem Stability:** NEGATIVE (-) — High erosion degrades habitat, silts waterways, and destroys the foundation that the ecosystem depends on.

Key Discoveries

- Removing apex predators causes herbivore populations to explode without natural checks
- Unchecked herbivores overgraze vegetation, leaving soil bare and vulnerable
- Vegetation loss triggers erosion that damages the entire ecosystem — even rivers and streams
- Restoring predators can reverse the cascade, but recovery takes years or decades
- Wolves in Yellowstone literally changed the course of rivers by allowing vegetation to regrow on banks

Multiple Choice Post-Assessment Answer Key

1. In your trophic cascade model, removing the Apex Predator Population caused Erosion Rate to spike. Which chain of cause-and-effect relationships from the model explains this?

Correct: B) No predators led to herbivore explosion, which overgrazed vegetation, which exposed bare soil to erosion

Correct! The model traces a clear causal chain: predator removal leads to unchecked herbivore growth, which leads to overgrazing and vegetation decline, which leads to bare soil vulnerable to wind and water erosion. This is a trophic cascade.

2. Your model shows that after predator reintroduction at 50%, ecosystem recovery took significantly longer than the original collapse. What does this asymmetry demonstrate?

Correct: B) Degraded ecosystems can become trapped in unhealthy stable states where damage is self-reinforcing, making recovery slower than collapse

Correct! Ecosystem degradation can be self-reinforcing. Bare, eroded soil cannot support new plant growth, which keeps herbivores concentrated on remaining vegetation, preventing recovery. Breaking out of this degraded state requires more time than the original collapse.

3. A student argues that removing wolves from a park should only affect elk, since wolves only eat elk. Using your model, which evidence contradicts this claim?

Correct: B) Removing predators triggers a cascade through herbivore, vegetation, erosion, and ecosystem stability components, affecting the entire system

Correct! The model shows that changes at one trophic level cascade through the entire system. Wolves affect elk, elk affect vegetation, vegetation affects erosion, and erosion affects ecosystem stability. Four components change from removing just one.

4. In the model, Vegetation Health has a POSITIVE relationship with Ecosystem Stability and a NEGATIVE relationship with Erosion Rate. If Vegetation Health drops to 20%, what does the model predict?

Correct: C) Erosion Rate increases dramatically while Ecosystem Stability declines, creating compounding damage

Correct! Low vegetation means less root structure to hold soil (higher erosion) AND less habitat and food for diverse species (lower ecosystem stability). These effects compound each other, accelerating ecosystem decline.

5. Based on your model data, a wildlife manager must choose between two strategies for a park with no predators. Which strategy does the model evidence best support?

Correct: B) Reintroduce apex predators to restore top-down control, which will cascade through herbivores, vegetation, and erosion over time

Correct! The model shows that the root cause of the cascade is predator loss. Reintroducing predators addresses the source of the problem, restoring the top-down control mechanism that regulates the entire cascade chain. Treating downstream symptoms alone does not fix the underlying cause.

7 Collaborative Learning Protocols

ModelIt!

Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

ModelIt!

Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

ModelIt!

Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Predators are bad for ecosystems"	Apex predators are essential regulators. Without them, herbivore populations grow unchecked and damage the very habitat that supports all species. Predators maintain balance through top-down control.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Removing one species only affects that species"	Ecosystems are interconnected networks. Removing one species — especially a keystone species like an apex predator — triggers cascading effects through multiple trophic levels that can transform entire landscapes.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Ecosystems always recover naturally"	Degraded ecosystems can become trapped in unhealthy stable states. Without intervention, overgrazing prevents vegetation recovery, which prevents erosion reduction, which prevents ecosystem recovery. Sometimes the system needs an external push to restart.